

LZW Question Generator Usability and Visual Perception Questionnaire

* Required

Briefing

The aim of this experiment is to evaluate the suitability, usability, and perceived effectiveness of an LZW question generator. In particular, the study investigates how users experience the system and how automatically generated variations, such as controlled symbol probability distributions (i.e. varying how frequently the same symbols appear) and input length, influence perceived difficulty and fairness. User involvement is essential because usability and perceived difficulty cannot be accurately assessed without observing real users interacting with the system.

Your feedback will help determine whether the generator functions as intended and whether the generated questions are clear, intuitive, and appropriately challenging. During this session, you will be given time to browse and use the question generator.

You will generate and complete LZW compression and decompression questions using the system. Observation will focus solely on how the system is used (for example, question generation, interaction flow, and any visible difficulties encountered). Participation assumes a basic understanding of how to complete LZW compression and decompression questions.

After using the system, you will be asked to complete a short questionnaire about your experience. The questions will focus on usability, perceived difficulty, fairness, and any differences you noticed between generated question types. You are free to continue interacting with the system while completing the questionnaire if you wish.

It is important to emphasise that this evaluation concerns the system, not you. There are no right or wrong behaviours during this study. You are free to withdraw from the experiment at any point without giving a reason.

1. Do you agree to taking part in this evaluation? *

☐ Yes

☐ No

Survey

2. How did you find using the question generator? *

- ☐ Very Intuitive
- ☐ Intuitive
- ☐ Neutral
- ☐ Unintuitive
- ☐ Very Unintuitive

3. Did you trust the generator to produce fair questions? *

- ☐ Yes
- ☐ No
- ☐ Don't Know

4. Did questions with different symbol patterns feel different in difficulty, even when they required the same number of steps to solve? *

- ☐ Yes
- ☐ No
- ☐ Maybe
- ☐ Don't know

5. How did the difficulty of questions with repetitive symbol patterns compare to questions with evenly distributed symbols? *

- ☐ Much more difficult
- ☐ Slightly more difficult
- ☐ About the same difficulty
- ☐ Slightly easier
- ☐ Much easier

6. How much do you agree with the following statement?

Differences in automatically generated input generated can impact perceived difficulty. *

- ☐ Strongly agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

7. How much do you agree with the following statement?

As the length of the string increased, the number of steps required to solve the problem increased proportionally. *

- ☐ Strongly Agree
- ☐ Agree
- ☐ Neutral
- ☐ Disagree
- ☐ Strongly disagree

8. Do you have any other thoughts after using the generator?

Debriefing

The main aim of this experiment was to investigate the suitability, usability, and perceived effectiveness of the LZW question generator. In particular, the study examined how users interacted with the system and whether automatically generated variations influenced perceived difficulty and fairness. More specifically, I was observing how you navigated the interface, how you generated and approached compression and decompression questions, and whether any aspects of the system appeared confusing or unintuitive. The study also aimed to identify whether differences in automatically generated inputs affected your perception of difficulty, even when the number of required solution steps was similar.

The data collected during this session consists of observational notes regarding system interaction and your responses to the post-task questionnaire. No aspect of this study evaluated your individual performance or ability; the focus was entirely on assessing the system.

If you have any comments, concerns, or questions about the experiment, you are welcome to raise them now. Please take note of my email address (2768771K@student.gla.ac.uk) and the module co-ordinator's email address (Oana.Andrei@glasgow.ac.uk).

If you have any further questions or concerns about the study at a later time, please feel free to contact us. Thank you very much for your participation, your feedback is greatly appreciated!

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